Activity 06: Case Analysis – Avoiding Misleading Proportions

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Course: DSA0601- Data Handling and Visualization

Question 1: Truncated Y-Axis in Sales Growth

A retail company presents the following monthly sales (in ₹ lakhs):

Month Sales

January 95

February 97

March 99

April 101

May 103

June 105

The marketing team creates a bar chart with the Y-axis starting at 90 instead of 0. The chart visually suggests that June sales are almost twice those of January.

Tasks:

1. **Explain why the chart is misleading.**

Ans: The chart is misleading because the Y-axis starts at 90 instead of 0. This exaggerates the small increase in sales, making June sales appear almost twice as high as January sales. In reality, sales increased only from ₹95 lakhs to ₹105 lakhs, which is about a 10.5% increase.

2. **Identify the visualization principle that has been violated.**

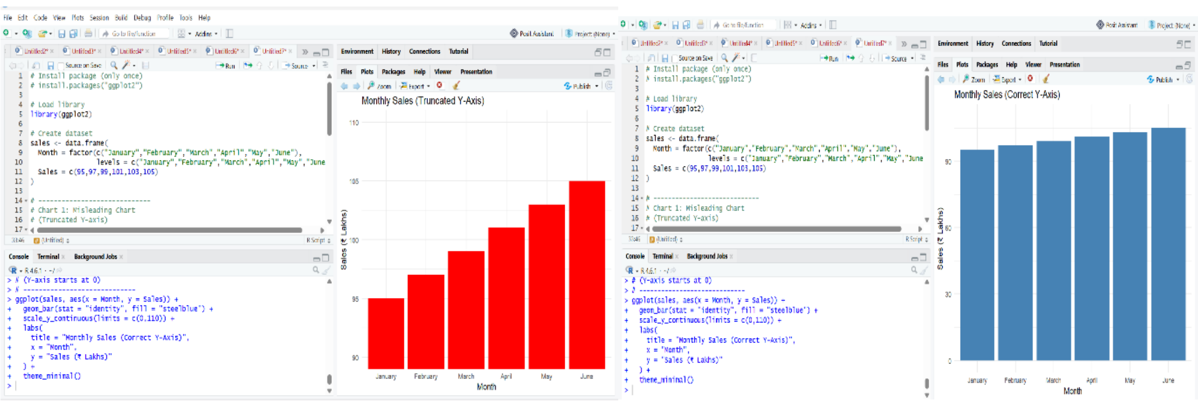
Ans: The principle of maintaining an appropriate and honest scale has been violated. Bar charts should generally start from zero so that the bar lengths accurately represent the data values without exaggeration

3. **Recommend an improved visualization method.**

Ans: A bar chart with the Y-axis starting at zero or a simple line chart should be used. Both methods display the gradual increase in sales accurately without misleading viewers.

4. **Describe how this misleading chart could influence business decisions.**

Ans: Managers may believe sales have grown significantly and make unnecessary decisions such as increasing production, expanding inventory, or investing heavily in marketing. This could result in poor financial planning and inefficient use of resources.



Question 2: Pie Chart with Incorrect Proportions

A university reports student enrollment in four departments.

Department Students

Computer Science 420

Electronics 280

Mechanical 200

Civil 100

The designer accidentally creates a pie chart where the slices occupy approximately **50%, 30%, 15%, and 15%** instead of the correct proportions.

Tasks:

1. Calculate the correct percentage for each department.

Ans: Total students = 420 + 280 + 200 + 100 = 1000

Department	**Students**	**Percentage**
Computer Science	420	42%
Electronics	280	28%
Mechanical	200	20%
Civil	100	10%

2. **Explain why the displayed chart is misleading.**

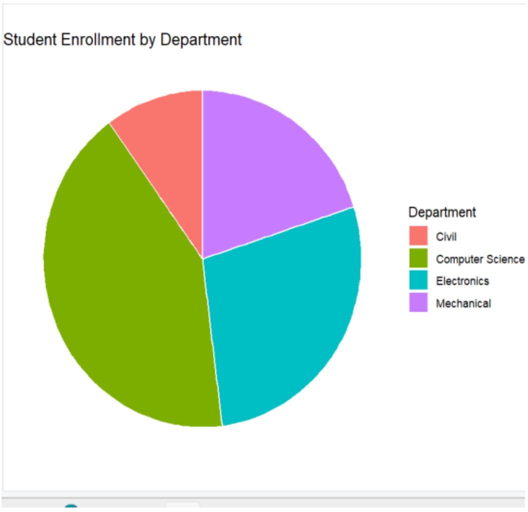
Ans: The displayed pie chart uses incorrect slice sizes of 50%, 30%, 15%, and 15%, which do not match the actual data. This gives a false impression of the student distribution, especially by overstating Computer Science and Civil enrolments.

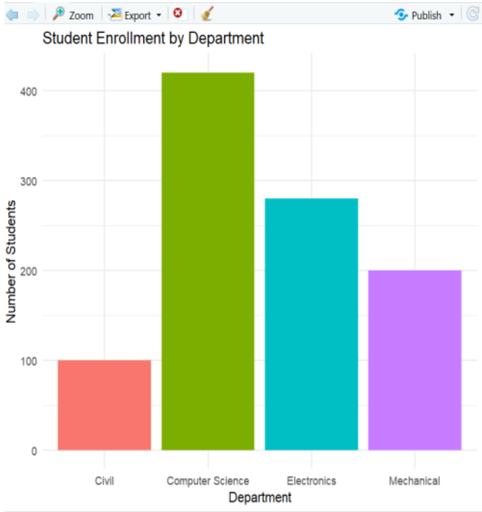
3. **Suggest an alternative chart that would improve comparison accuracy.**

Ans: A bar chart is a better choice because it allows viewers to compare department enrolments more accurately using aligned bar lengths.

4. **Discuss how such errors could affect institutional planning.**

Ans: Incorrect enrolment figures could lead to poor allocation of faculty, classrooms, laboratory facilities, and budgets. Departments may receive more or fewer resources than actually required.





Question 3: Unequal Bar Widths in Revenue Comparison

A software company compares annual revenue.

Product Revenue (₹ Crores)

A 60

B 75

C 90

D 105

Instead of using equal-width bars, the designer creates bars with increasing widths, making Product D appear nearly **three times larger** than Product A.

Tasks:

1. **Identify the graphical error.**

Ans: The graphical error is the use of unequal bar widths. Bar charts should use equal-width bars so that only the height represents the data values.

2. **Explain how unequal bar widths distort perception.**

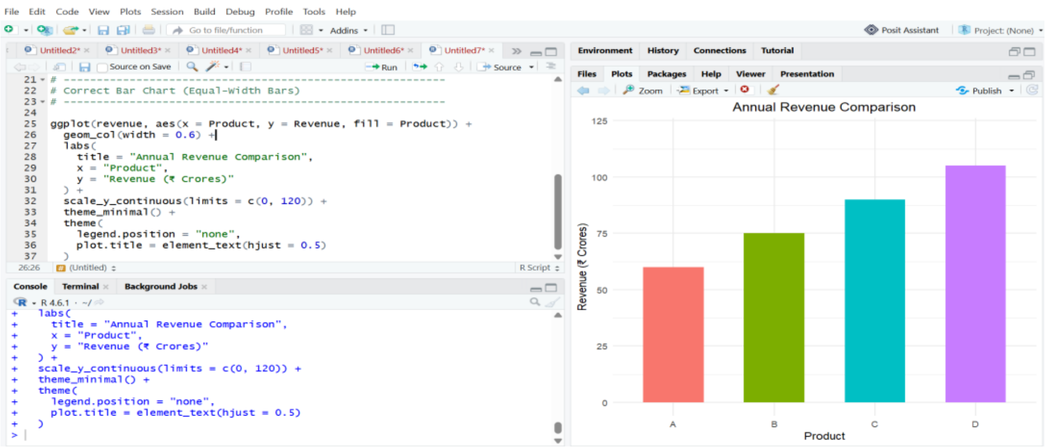
Ans: Increasing the width of bars increases their overall area, making larger bars appear much more significant than they actually are. This causes viewers to overestimate the revenue differences between products.

3. **Redesign the visualization by specifying the correct design principles.**

Ans: The redesigned chart should use equal-width bars, a consistent spacing between bars, a Y-axis starting at zero, clear labels, and a simple two-dimensional layout without unnecessary visual effects.

4. **Discuss the possible consequences if investors rely on this chart.**

Ans: Investors may incorrectly assume Product D is far more profitable than Product A and make poor investment decisions. This could affect company valuation, investment priorities, and shareholder confidence.



Question 4: 3D Column Chart Distortion

An energy company reports renewable energy generation.

Source Energy (GWh)

Solar 450

Source Energy (GWh)

Wind 500

Hydro 550

Biomass 520

The analyst uses a 3D column chart with perspective effects. Due to the viewing angle, the Hydro bar appears almost 40% larger than Wind, although the actual difference is only 10%.

Tasks:

1. **Explain why 3D charts can create misleading proportions.**

Ans: Three-dimensional charts introduce perspective effects that make some columns appear taller or larger than they actually are. The viewing angle can distort the visual comparison between values.

2. **Compare the perceived difference with the actual difference.**

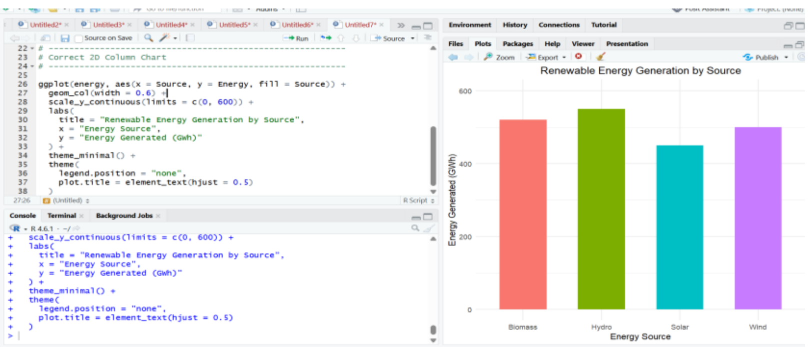
Ans: Wind generates 500 GWh and Hydro generates 550 GWh. The actual increase is 50 GWh, which is 10%. However, the 3D chart makes Hydro appear about 40% larger, greatly exaggerating the difference.

3. **Recommend a better visualization.**

Ans: A simple two-dimensional bar chart or column chart with a zero baseline provides an accurate comparison of energy generation across different renewable sources.

4. **Discuss the importance of avoiding visual distortion in sustainability reporting.**

Ans: Accurate sustainability reports help governments, investors, and the public make informed decisions. Misleading charts may create false impressions about renewable energy performance, leading to incorrect policy decisions and reduced public trust.



Question 5: Bubble Chart with Incorrect Scaling

A healthcare organization compares disease cases across five districts.

District Cases

A 150

B 250

C 350

D 450

E 550

The analyst scales the bubble radius directly to the number of cases rather than scaling the bubble area. As a result, District E appears more than ten times larger than District A.

Tasks:

1. **Explain why scaling the radius instead of the area is misleading.**

Ans: When the radius is scaled directly, the bubble area increases much faster because area is proportional to the square of the radius. This makes larger values appear much larger than they actually are and exaggerates the differences.

2. **Describe the correct method for proportional bubble sizing.**

Ans: Bubble charts should scale the area of each bubble according to the data values. The radius should be calculated from the square root of the value so that the bubble areas accurately represent the underlying data.

3. **Suggest an alternative visualization suitable for this dataset.**

Ans: A bar chart is the most suitable alternative because it provides a simple and accurate comparison of disease cases across the five districts without introducing area-based perception errors.

4. **Discuss the potential impact of this visualization on public health resource**

allocation.

Ans: An exaggerated bubble chart may lead health officials to allocate excessive resources to one district while underfunding others. This can reduce the efficiency of healthcare services, affect disease control measures, and result in unequal distribution of medical resources.